

Question number	Answer	Additional guidance	Mark
5(a)	An explanation that makes reference to the following points: - because platelets {contain / release} thromboplastin (1) - thromboplastin results in the formation of thrombin from prothrombin (1)	IGNORE descriptions of plaque formation prior to release of thromboplastin by platelets IGNORE endothelial cells	(2)

Question number	Answer	Additional guidance	Mark
5(b)	person bleeds {too easily / uncontrollably / internally / excessively} (1)	Examiners will need to check any other risks but not accept just side effects or symptoms ACCEPT (severe / frequent) bruising (prolonged / frequent) nose bleeds heavy periods (in women) liver damage kidney damage IGNORE named side effects	(1)

Question number	Answer	Mark
5(c)(i)	The only correct answer is D <i>A is incorrect because there are 96 bases for 32 amino acids plus a stop codon</i> <i>B is incorrect because there are 96 bases for 32 amino acids plus a stop codon</i> <i>C is incorrect because there are 96 bases for 32 amino acids plus a stop codon</i>	(1)

Question number	Answer	Additional guidance	Mark
5(c)(ii)	An explanation that makes reference to the following points: - because (if the active site is blocked) thrombin will not be able to convert fibrinogen into fibrin (1) - without fibrin, {platelets / blood cells} will not get trapped (forming a blood clot) (1)	NB must be an idea of what will not happen for 2 marks to be awarded ACCEPT less ACCEPT without fibrin, {mesh won't form / less mesh will form}	(2)

Question number	Answer	Additional guidance	Mark
5(c)(iii)	An explanation that makes reference to two of the following points: • variegins bind to the <u>active site</u> of thrombin / an enzyme substrate complex is formed / thrombin lowers the activation energy (1) • therefore the peptide bonds are broken (by thrombin) (1) • by hydrolysis / by a hydrolysis reaction (1)	ACCEPT ESC for enzyme substrate complex	(2)

Question number	Answer	Additional guidance	Mark
5(c)(iv)	An answer that makes reference to two of the following points: • (overall) variegain results in less (mass) blood clot than {drug H / the other anticoagulant} (1) • there is less variation in the size of the blood clot (in patients treated) with variegain (1) • cannot comment on the effectiveness of {drugs / variegain} compared with no drug as no control (1) • cannot comment on {significance / validity} of data as no statistical analysis (1)	ACCEPT converse throughout ACCEPT reduces blood clot compared to drug H prevents blood clot better than drug H more effective than drug H (in reducing blood clot) ACCEPT smaller range ACCEPT cannot comment on effectiveness of drug H as no control ACCEPT no mean <u>and</u> error bars shown not <u>valid</u> as small sample size	(2)